Linear Equation in Two Variables

- Recap:

Equation

$f(x) = 0$

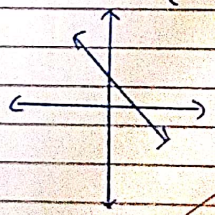
LHS ← variable → RHS

Linear Eq.
$f(x) = 0$
$2x + 1 = 0$

Quadratic Eq.
$ax^2 + bx + c = 0$
where $a \neq 0$

Linear Equation in two variable (x, y)

$ax + by + c = 0$ (where a, b, c are constants)
$x, y \rightarrow$ variables



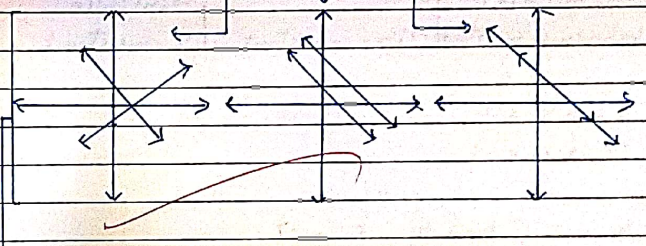
If $a = 0$
$0x + by + c = 0$
$by + c = 0$

- Introduction:

$a_1x + b_1y + c_1 = 0$
$a_2x + b_2y + c_2 = 0$

$7x + 4y = 0$
$2x + 2s = 0$
} Not a linear equation

Two lines in a plane



Finding Solutions of a pair of linear equation in 2 variable

- Graphical
- Algebriac
  - Elimination Method
  - Substitution Method

→ word Problems ←

- **Algebraic Interpretation of pair of linear equation in 2 variables**

$a_1x + b_1y + c_1 = 0$

$a_2x + b_2y + c_2 = 0$

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
→ Intersecting	→ parallel	→ coinciding
→ no. of solu = 1	→ 0	→ infinite
→ behavior - consistant	→ in - consistant	→ consistant

$2x + 4y = 7$

$3x + 7y = -1$

$a_1 = 2, b_1 = 4, c_1 = -7$

$a_2 = 3, b_2 = 7, c_2 = 1$

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

result = intersecting

no. of solu = 1

$2x + 4y = 7$

$4x + 8y = 8$

$a_1 = 1, b_1 = 4, c_1 = 7$

$a_2 = 2, b_2 = 8, c_2 = 8$

$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

result = Parallel

no. of solu = 0

Exercise 3

1 - Sol: Let x and y be the numbers of boys and girls respectively.

The algebraic representation is is given by:

$x + y = 10$

$\Rightarrow x + y - 10 = 0$

$x + y = 4$

$\Rightarrow x - y + 2 = 0$

Graphical solution:

For $x + y - 10 = 0$

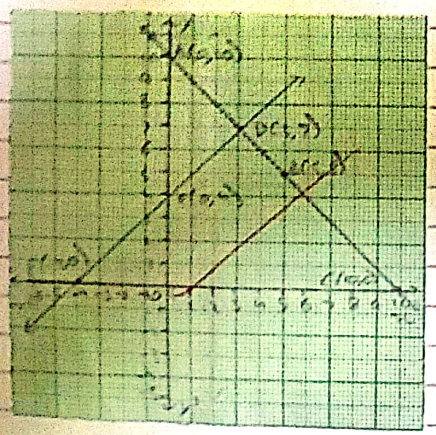
x	0	5	10
y	10	5	0

For $x - y + 4 = 0$

x	3	0	-4
y	7	4	0

Plot A(0,10), B(5,5) and C(10,0) and draw the line ABC

Plot D(3,7), B(0,4) and C(-4,0) and draw the line DEF.



The two lines intersect at D(3,7), which is the required graphic solution.

Hence, the no. of boys 3 and of girls is 7

5 - Sol: Let x and y be the cost of 1 pencil and 1 pen respectively.

The algebraic representation is is given by:

$5x + 7y = 50$

$\Rightarrow 5x + 7y - 50 = 0$

$7x + 5y = 46$

$\Rightarrow 7x + 5y - 46 = 0$

Graphical solution:

x	2	3	10
y	$40/7$	5	0

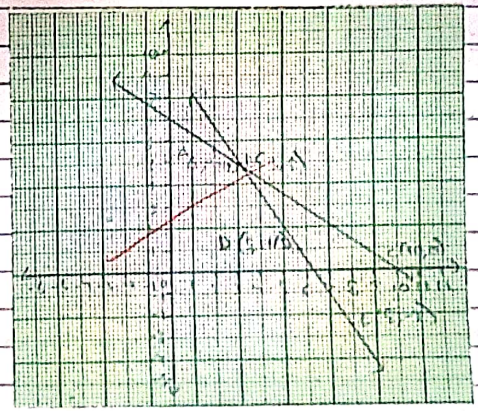
x	3	5	8
y	5	$1/5$	-2

For $5x + 7y - 50 = 0$

For $7x + 5y - 46 = 0$

Plot A(2, 40/7), B(3,5) and C(10,0) and draw line ABC.

Plot D(3,5), E(5, 11/5) and F(8,-2) and draw line BDE.



Scanned with OKEN Scanner

The two lines intersect at B(3,5), which is the required graphic solution.

Hence, the cost of 1 pen is ₹35 and cost of 1 pencil is ₹3.

2-i-Sol) The given lines are $5x-4y+8=0$ and $7x+6y-9=0$

Here, $\frac{a_1}{a_2} = \frac{5}{7}$, $\frac{b_1}{b_2} = \frac{-4}{6} = \frac{-2}{3}$ and $\frac{c_1}{c_2} = \frac{8}{-9} = \frac{-8}{9}$

Thus, $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

Hence, the given lines intersect at a point.

ii-Sol) The given lines are $9x+3y+12=0$ and $18x+6y+24=0$

Here, $\frac{a_1}{a_2} = \frac{9}{18} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{3}{6} = \frac{1}{2}$ and $\frac{c_1}{c_2} = \frac{12}{24} = \frac{1}{2}$

Thus, $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Hence, the given lines are coincident.

iii-Sol) The given lines are $6x-3y+10=0$ and $2x-y+9=0$

Here, $\frac{a_1}{a_2} = \frac{6}{2} = 3$, $\frac{b_1}{b_2} = \frac{-3}{-1} = 3$ and $\frac{c_1}{c_2} = \frac{10}{9}$

Thus, $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

Hence, the given lines are coincident.

3-i-Sol) The given linear equations are:

$3x+2y-5=0$ and $2x-3y-7=0$

Here, $\frac{a_1}{a_2} = \frac{3}{2}$ and $\frac{b_1}{b_2} = \frac{2}{-3} = -\frac{2}{3}$

Clearly $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

Hence, the given pair of linear equations are consistent.

ii-Sol) The given linear equation can be written as:

$2x-4y-8=0$

and $4x-6y-9=0$

Here, $\frac{a_1}{a_2} = \frac{2}{4} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{-4}{-6} = \frac{2}{3}$

and $\frac{c_1}{c_2} = \frac{-8}{-9} = \frac{8}{9}$.

Thus, $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

Hence, the given pair of linear equations is inconsistent.

iii-Sol) The given linear equations can be written as:

$\frac{3}{2}x + \frac{5}{2}y - 7 = 0$

and $9x - 10y - 14 = 0$

Here, $\frac{a_1}{a_2} = \frac{3/2}{9} = \frac{1}{6}$, $\frac{b_1}{b_2} = \frac{5/2}{-10} = -\frac{1}{6}$

Thus, $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

Hence, the given pair of linear equations is consistent.

Scanned with OKEN Scanner

4-i-sol) The given linear equations can be written as:

$x+y-5=0$

and $2x+2y-10=0$

Here, $\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, and $\frac{c_1}{c_2} = \frac{-5}{-10}$

Hence, the given linear equations are consistent.

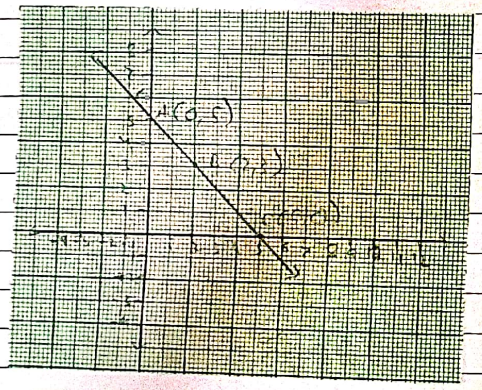
Graphical solutions:

For $x+y-5=0$

x	0	2	5
y	5	3	0

For $2x+2y-10=0$

x	0	2	5
y	5	3	0



Hence, the given systems of equations is consistent having an infinite number of solutions.

5-sol) Let 'x m' and 'y m' be the length and width of the garden.

By the question,

$x = y + 4$

$\Rightarrow x - y - 4 = 0$

and $\frac{1}{2}(2x+2y) = 36$

$\Rightarrow x + y - 36 = 0$

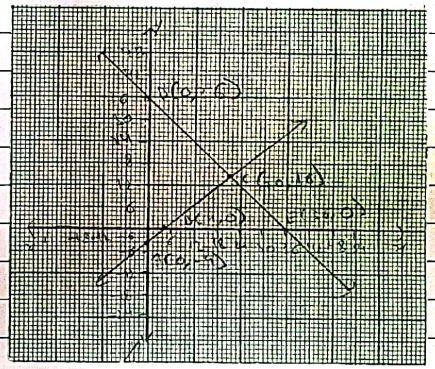
Graphical Solution:

For $x-y-4=0$

x	0	4	20
y	-4	0	16

For $x+y-36=0$

x	0	20	36
y	36	16	0



Hence, $x = 20$ m and $y = 16$ m are the dimensions of the garden.

Exercise - 3.2

1-i-sol) The given equations are $x+y=14$ — (1)
and $x-y=4$ — (2)
From (2) $x=4+y$ — (3)
Putting the value of x in (1),
$4+y+y=14$
$2y=10$
$y=5$
Putting the value of y in (3), $x=5+4$
$=9$
Hence, $x=9$ and $y=5$

ii-sol) The given equations are $s+t=11$ — (1)
and $\frac{s}{3}+\frac{t}{2}=6$ — (2)
From (1), $s=3+t$ — (3)
Putting the value of s in (2),
$\frac{t+3}{3}+\frac{t}{2}=6$
$2(3+t)+3t=36$
$6+2t+3t=36$
$5t=30$
$t=6$
Putting the value of t in (3),
$s=3+6$
$=9$
Hence, $s=9$ and $t=6$

2-Sol) The given equations are $2x+3y=11$ — (1)
and $3x-4y=-24$ — (2)
From (1), $x=\frac{11-3y}{2}$ — (3)
Putting the value of x in (2),
$11-3y-4y=-24$
$11-7y=-24$
$11+24=7y$
$7y=35$
$y=5$
Putting the value of $5y$ in (3),
$x=\frac{11-3(5)}{2}$
$x=\frac{11-15}{2}$
$x=\frac{-4}{2}$
$x=-2$
Hence, $x=-2$ and $y=5$
Now $y=mx+3$
$5=m(-2)+3$
$5=-2m+3$
$2m=-2$
Hence, $m=-1$

3-i-solu → Let 'x' and 'y' be two numbers, where x>y.

By the question, $x - y = 26$ — (1)

and $x = 3y$ — (2)

Putting the value of x in (1),

$3y - y = 26$

$2y = 26$

$y = 13$

Putting the value of y in (2),

$x = 3(13)$

$x = 39$

Hence, the required numbers are 39 and 13.

ii-solu Let 'x' and 'y' be two supplementary angles, where x>y.

By the question, $x + y = 180$ — (1)

and $x - y = 18$ — (2)

Putting the value of $x = (18 + y)$ in (1),

$18 + y + y = 180$

$2y = 162$

$y = 81$

Putting value of y in (2),

$x - 81 = 18$

$x = 18 + 81$

$x = 99$

Hence, required angles are 99° and 81°.

Exercise - 3.3

1-i-solu The given equations are:

$x + y = 5$ — (1)

and $2x - 3y = 4$ — (2)

Elimination method:

Multiplying (1) by 2:

$2x + 2y = 10$ — (3)

Subtracting (2) from (3),

$5y = 6$

$y = \frac{6}{5}$

Putting value of y in (1)

$x + \frac{6}{5} = 5$

$x = 5 - \frac{6}{5}$

$x = \frac{25 - 6}{5}$

$x = \frac{19}{5}$

Hence, $x = \frac{19}{5}$ and $y = \frac{6}{5}$

Substitution Method:

From (2), $2x = 4+3y$

$\Rightarrow x = \frac{4+3y}{2}$ — (4)

Substituting this value of x in (1), we get:

$\frac{4+3y}{2} + y = 5$

$\Rightarrow 4+3y+2y = 10$

$\Rightarrow 4+5y = 10$

$\Rightarrow y = \frac{6}{5}$

Substituting this value of y in (4), we get:

$x = \frac{4+3(\frac{6}{5})}{2}$

$x = \frac{4+\frac{18}{5}}{2} = \frac{20+18}{10} = \frac{38}{10} = \frac{19}{5}$

Hence, $x = \frac{19}{5}$ and $y = \frac{6}{5}$.

The given equations are:

$3x+4y = 10$ — (1)

and $2x-2y = 2$ — (2)

Elimination method:

Multiplying (1) by 2,

$4x - 4y = 4$ — (3)

Adding (1) and (3),

$7x = 14$

$x = 2$

Putting $x = 2$ in (2),

$2(2) - 2y = 2$

$4 - 2y = 2$

$-2y = -2$

$y = 1$

Hence, $x = 2$ and $y = 1$

Substitution Method:

From (2), $2x = 2+2y$

$x = y+1$ — (4)

Substituting this value of x in (1),

$3(y+1)+4y = 10$

$3y+3+4y = 10$

$7y = 7$

$y = 1$

Substituting this value of y in (4),

$x = 1+1 = 2$

Hence, $x = 2$ and $y = 1$

2-i-sol) Let $\frac{x}{y}$ be the fraction.

By the question, $\frac{x+1}{y-1} = 1$ and $\frac{x}{y+1} = \frac{1}{2}$

$x+1 = y-1$

$x-y+2 = 0$ — (1)

and $2x = y+1$

$2x-y-1 = 0$ — (2)

Subtracting (1) from (2),

$x-3 = 0$

$x = 3$

Scanned with OKEN Scanner

anyone takes the book for 10 days
then he/she has to pay fixed charges
for 3 days and additional charges for 7 days. Sarita takes a book from Library for 7 days and paid 27 Rs and Babita take a book from Library for 5 days and paid 21 Rs.

(a) If fixed charges for the library for the first 3 days is x Rs and additional charges for each day is y Rs/day then pair of linear equation satisfying the Sarita case is:

(i) $3x + 4y = 27$ (ii) $x + 4y = 27$ (iii) $x - 4y = 27$ (iv) $x + 7y = 27$

(b) If fixed charges for the library for first 3 days is x Rs and additional charges for each is y Rs/day then pair of linear equation satisfying Babita case is:

(i) $x + 5y = 21$ (ii) $3x + 2y = 21$ (iii) $x - 5y = 21$ (iv) $x + 2y = 21$

(c) Fixed charges and additional charges of the library is:

(i) fixed = 5 Rs and additional = 10 Rs/day (ii) fixed = 15 Rs and additional = 3 Rs/day

(iii) fixed = 1 Rs and additional = 5 Rs/day (iv) fixed = 10 Rs and additional = 5 Rs/day

(d) If Sarita keeps the book for 15 days, then how much she has to pay to the library.

(i) 49 Rs (ii) 50 Rs (iii) 51 Rs (iv) 20 Rs

(e) If Babita keeps the book for 10 days, then how much she has to pay to the library.

(i) 36 Rs (ii) 30 Rs (iii) 25 Rs (iv) 32 Rs

Scanned with OKEN Scanner